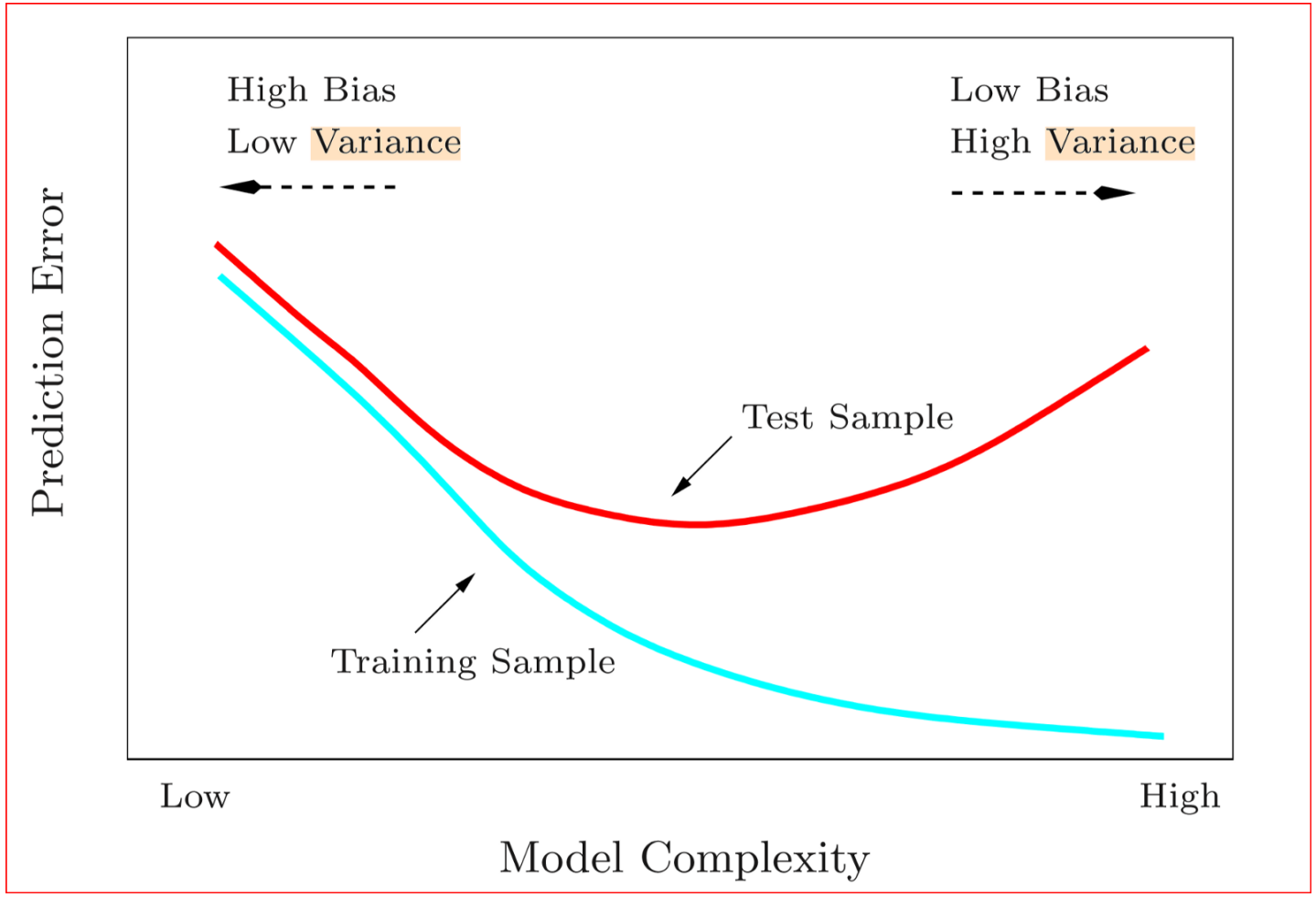
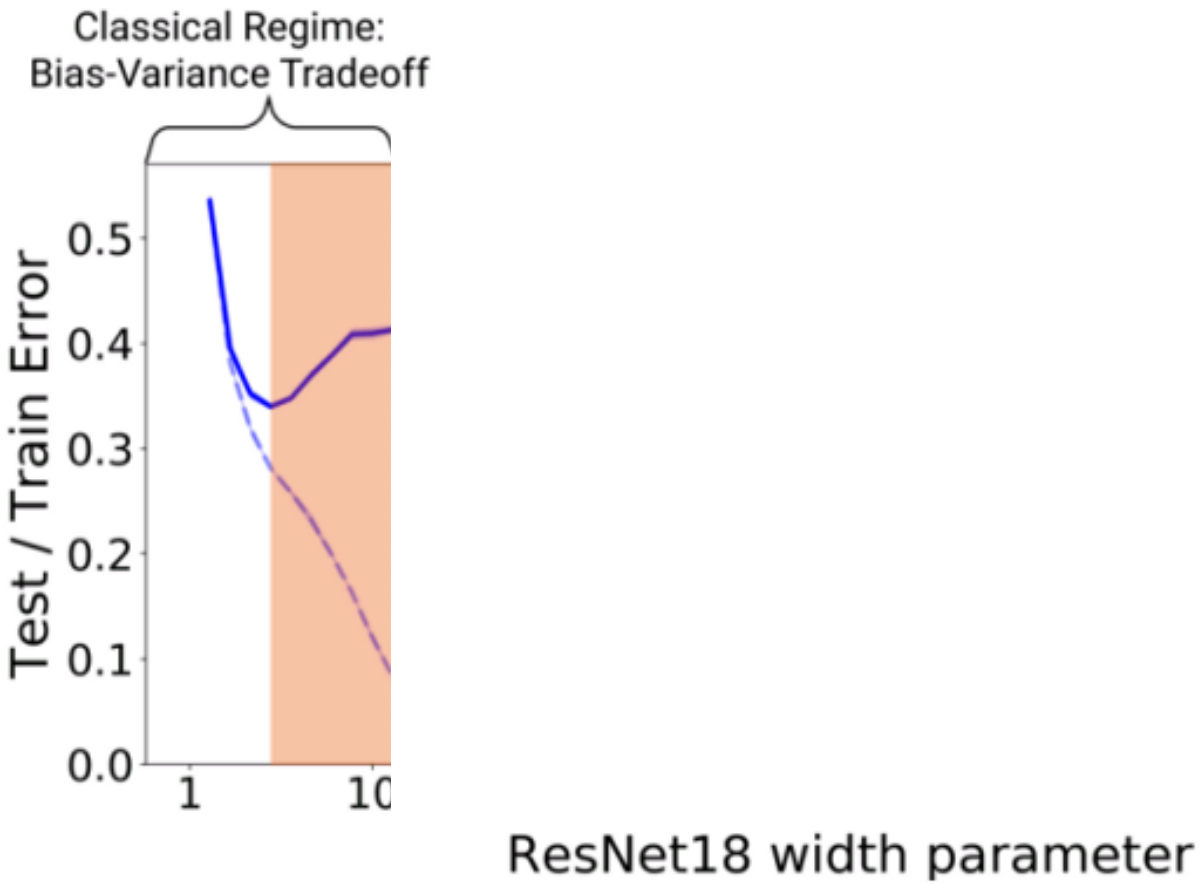
Case 1: Bias-Variance trade-off



[source: https://storybydata.com/datacated-challenge/bias-variance-dilemma/]

Case 1: Bias-Variance trade-off

Increasing a deep network's parameters



CIFAR-10 with 15% label noise

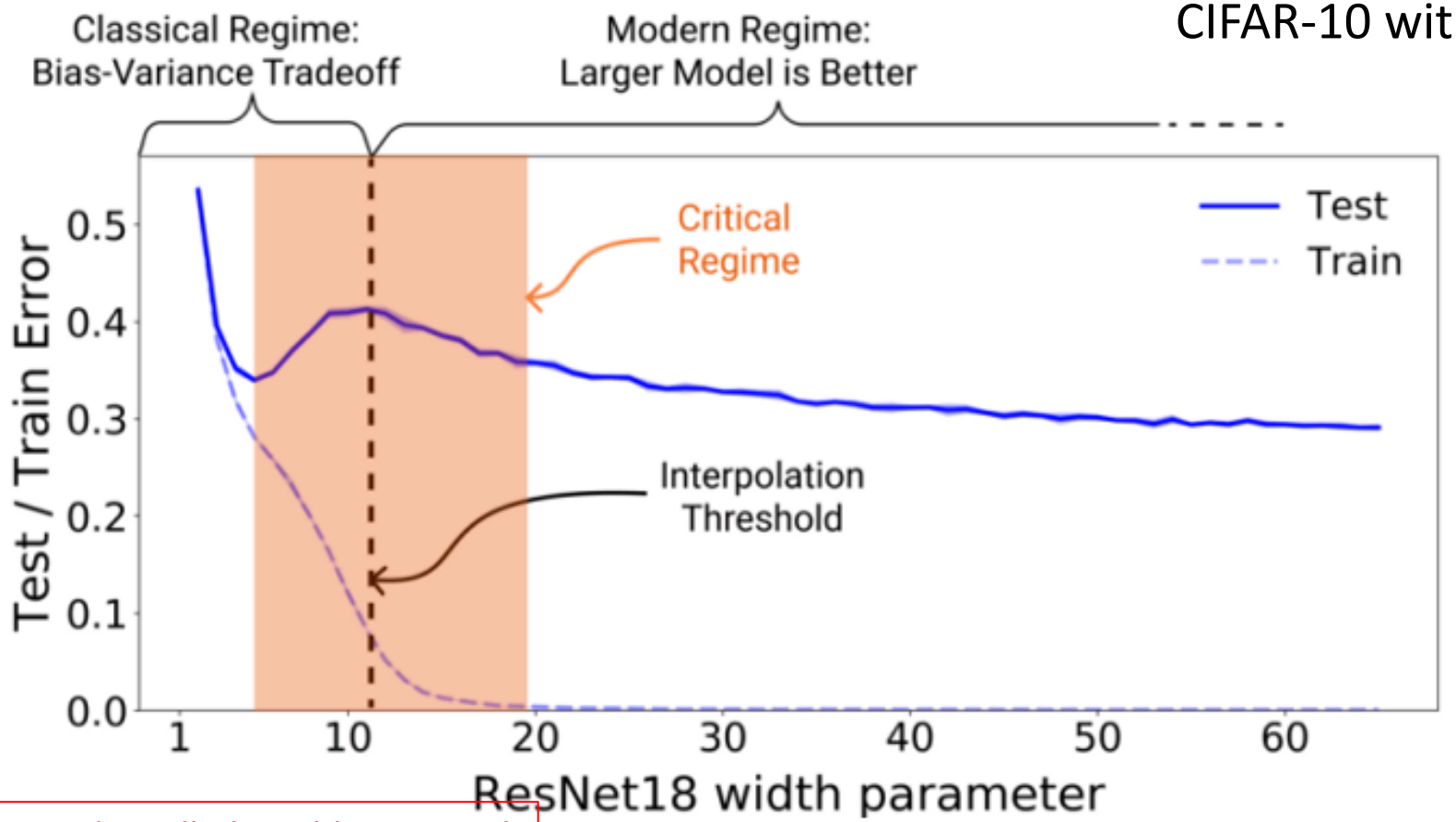
Test
Train

[Nakkiran, "DEEP DOUBLE DESCENT: WHERE BIGGER MODELS AND MORE DATA HURT", ICLR 2020]
[Belkin et al., "Reconciling modern machine-learning practice and the classical bias–variance trade-off", PNAS 2020]
[Geiger et al. "The jamming transition as a paradigm to understand the loss landscape of deep neural networks", 2018]

Case 1: Bias-Variance trade-off and Double Descent

Increasing a deep network's parameters

CIFAR-10 with 15% label noise



This is a "*new*" *phenomenon* and is called Double Descent!

[Nakkiran, "DEEP DOUBLE DESCENT: WHERE BIGGER MODELS AND MORE DATA HURT", ICLR 2020]

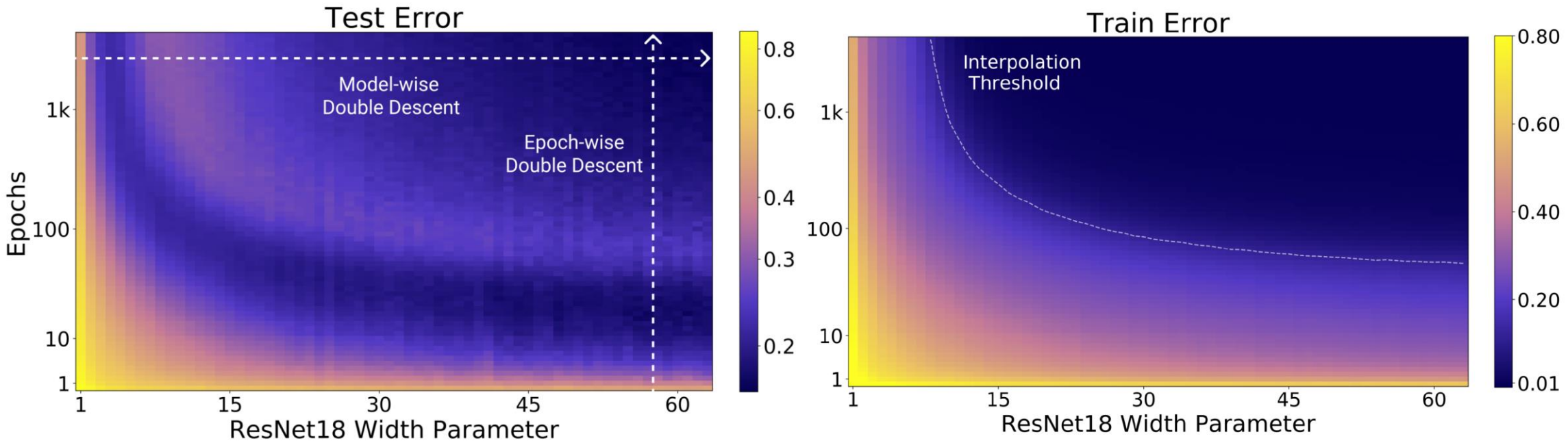
[Belkin et al., "Reconciling modern machine-learning practice and the classical bias–variance trade-off", PNAS 2020]

[Geiger et al. "The jamming transition as a paradigm to understand the loss landscape of deep neural networks", 2018]

Case 1: Bias-Variance trade-off and Double Descent

Increasing a deep network's parameters and training time

CIFAR-10 with 15% label noise

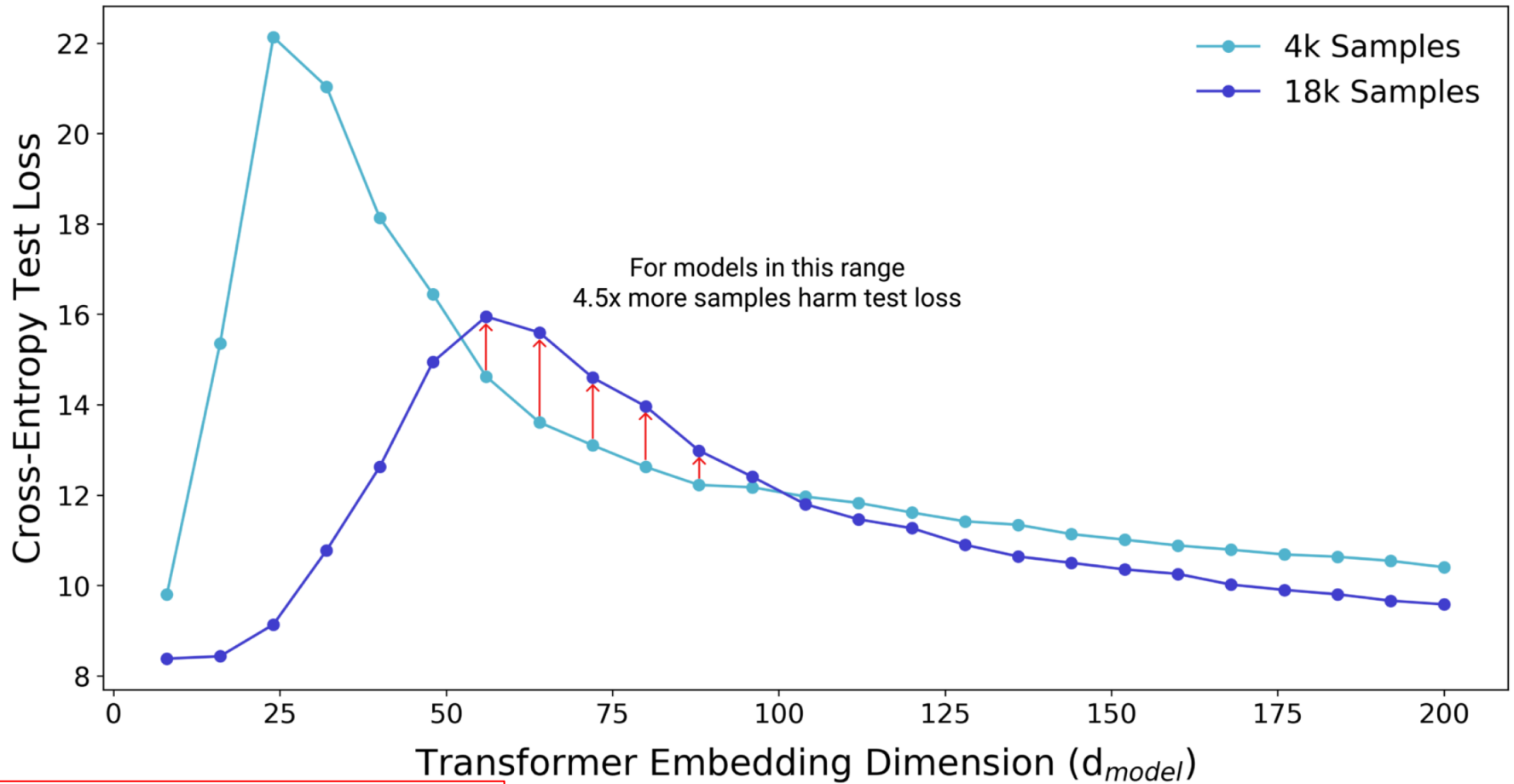


Similar phenomenon for the training time!

[Nakkiran, "DEEP DOUBLE DESCENT: WHERE BIGGER MODELS AND MORE DATA HURT", ICLR 2020]
[Belkin et al., "Reconciling modern machine-learning practice and the classical bias–variance trade-off", PNAS 2020]
[Geiger et al. "The jamming transition as a paradigm to understand the loss landscape of deep neural networks", 2018]

Case 1: Bias-Variance trade-off and Double Descent

Increasing a deep network's parameters and dataset size



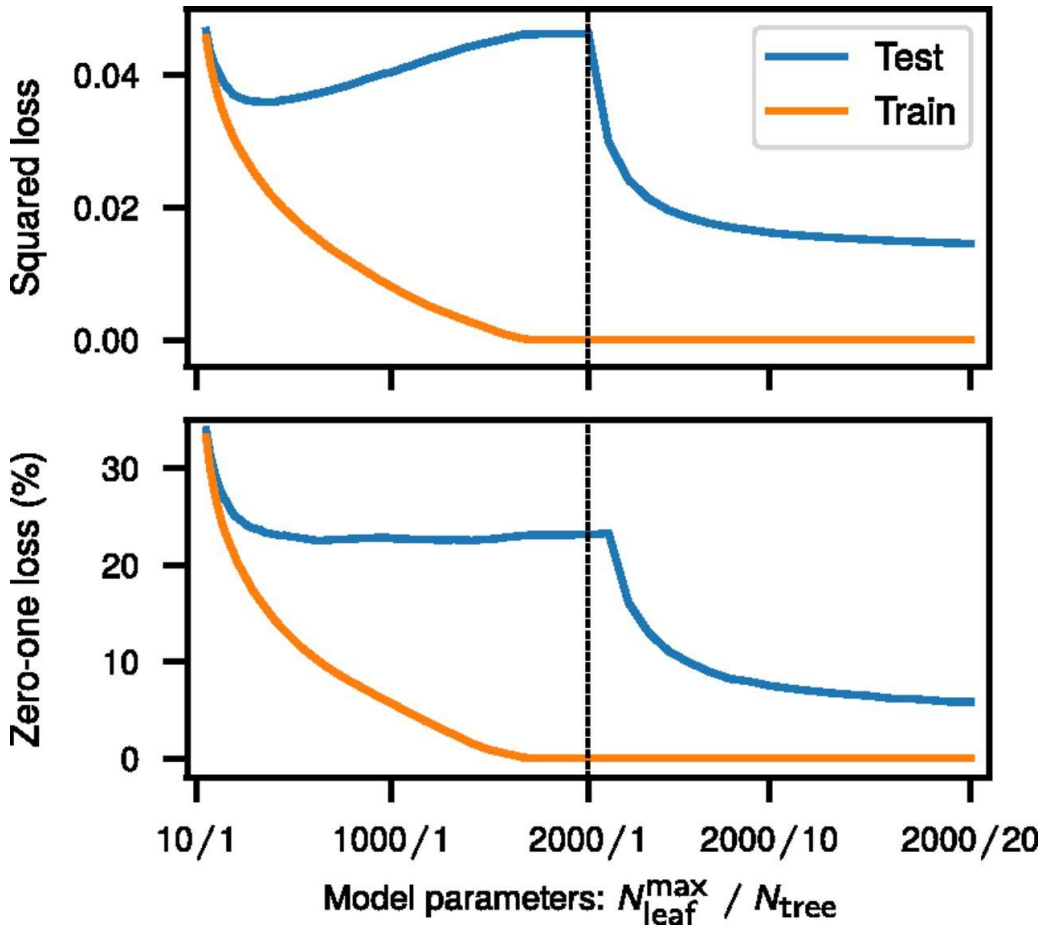
Transformer trained on IWSLT

Sometimes less (training data) can be more (useful)!

[Nakkiran, "DEEP DOUBLE DESCENT: WHERE BIGGER MODELS AND MORE DATA HURT", ICLR 2020]
[Belkin et al., "Reconciling modern machine-learning practice and the classical bias–variance trade-off", PNAS 2020]
[Geiger et al. "The jamming transition as a paradigm to understand the loss landscape of deep neural networks", 2018]

Case 1: Bias-Variance trade-off and Double Descent

Increasing a random forest's parameters



Not unique to deep learning!

[Nakkiran, "DEEP DOUBLE DESCENT: WHERE BIGGER MODELS AND MORE DATA HURT", ICLR 2020]

[Belkin et al., "Reconciling modern machine-learning practice and the classical bias–variance trade-off", PNAS 2020]

[Geiger et al. "The jamming transition as a paradigm to understand the loss landscape of deep neural networks", 2018]